

## Intuitive Overview of the Nash Equilibrium

How could you define what the Nash Equilibrium is? One way is to say that after any game, every player agrees that they're happy with their personal decision (given what everyone else did). That is, they made the best response to their opponent's best response.

Getting into a little philosophy, is this really the optimal outcome? So far, this is the best solution we have—that everybody does the best they could, assuming the opponent's best.

It is important to realize that the Nash Equilibrium doesn't necessarily mean both players have the best outcome. In some sense, it is the best outcome one can arrive at given that you also want stability. Thus, it can be thought of as the best outcome given the highest certainty. If certainty or stability is sacrificed, both players can sometimes settle on better outcomes.

Also, games don't necessarily have only one Nash Equilibrium. Sometimes they have two, sometimes they have none.

## Playing a Game

Suppose Luke and Burak are playing a game, where the outcomes of each move are listed in the table below:

|      |      | Burak |       |
|------|------|-------|-------|
|      |      | Left  | Right |
| Luke | Up   | 4, 8  | 7, 4  |
|      | Down | 6, 3  | 9, 2  |

How do we find the Nash Equilibrium? First, pretend you are Luke. You see that if Burak plays 'left', the best outcome is when you play 'down' (you get a 6). If Burak plays 'right', your best outcome is 'down'.

Now pretend you are Burak. If Luke plays 'up', you want to play 'left' (you get an 8). If Luke plays 'down', you want to play 'left'.

The Nash Equilibrium is the bottom left corner, (6,3), because both opponents prefer it despite what their opponent plays. This fits our definition of the Nash Equilibrium.

Notice that (6,3) isn't the best outcome for both players—(7,4) is. However, (7,4) isn't as stable as (6,3). If both players agree to choose (7,4) the entire game, one of the players could choose another square to get better results for themselves, which will necessarily be a worse outcome for their opponent!

## Another Game

An interesting way to define the Nash Equilibrium is as "a law no one wants to break even if there was no government enforcement". Think of when you're sitting at a traffic light. The values for each choice are modeled in the table below:

|          |      | Driver 2 |        |
|----------|------|----------|--------|
|          |      | Go       | Stop   |
| Driver 1 | Go   | -5, -5   | 1, 0   |
|          | Stop | 0, 1     | -1, -1 |

The discussion is similar to last time. If both players go, there will be an unpleasant collision. If both players stop, they will waste each other's time. Thus, performing the same analysis as in the last game, the two equilibriums in this example are when one player stops and the other goes! In fact, situations like this occur when a traffic light goes down.

## Math

A strategy profile  $s \in S$  is a Nash Equilibrium if and only if:

$$s_i \in BR_i(s_{-i})$$

Here  $s$  is a vector  $s = (s_1, s_2, \dots, s_n)$  that tells you what strategy each player uses.  $BR_i$  stands for 'best response' and means that each player's strategy is the best response to their opponent, as per our first definition.

More formally, a strategy profile  $s \in S$  is a Nash Equilibrium iff:

$$U_i(s_i, s_{-i}) \geq U_i(s'_i, s_{-i}) \quad \forall s'_i \in S_i \text{ players } i$$

Where:

- $U_i(s_i, s_{-i})$ : The payoff when the player performs  $s_i$  and the opponent performs  $s_{-i}$ .
- $U_i(s'_i, s_{-i})$ : The payoff of playing an alternative strategy given the opponent plays the same.